



Salaheddine STA

Software Engineer (C++) | Computer Vision | PhD in Computer Science

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LANGUAGES

French Fluent
English Professional
Arabic Native

SKILLS

Software Engineering

Software Design Performance Optimization
Software Validation Debugging Git CI/CD

Computer Vision

Camera Calibration Projective Geometry
2D/3D Pose Estimation Registration
Object Detection Image Segmentation
Point Cloud Processing(RGB-D) 3D Reconstruction

Artificial Intelligence

Deep Learning Machine Learning nnUNet
LLM NLP

Mathematical & Analytical Skills

Applied Mathematics Statistics Data Analysis
Linear Algebra Geometry

PROGRAMMING LANGUAGES & TOOLS

Programming

C++ (14/17/20) C Python Bash Rust

Libraries & Frameworks

Point Cloud Library (PCL) OpenCV
Insight Toolkit (ITK) Visualization Toolkit (VTK)
Vision-something-Libraries (VXL) CMake

Operating Systems

Linux Windows

PUBLICATIONS & PATENT

Patent Automatic determination of an appropriate positioning of a patient-specific instrumentation with a depth camera.
EP4082469B8 / US12290325B2. [Link](#) ↗

Thesis Towards a markerless Computer Assisted Orthopaedic Surgery system. [Link](#) ↗

Int. J. Medical Robotics 2021 Towards markerless computer assisted surgery. [Link](#) ↗

CAOS 2020 Towards markerless 3D pose estimation for CAOS: comparison study of depth cameras. [Link](#) ↗

CARS 2019 Towards a markerless Computer Assisted Orthopaedic Surgery system. [Link](#) ↗

PROFILE

Software Engineer with a PhD in Computer Science and over seven years of experience in software development, algorithm design, computer vision, and artificial intelligence. Specialized in designing high-performance software systems, 3D data processing, and integrating machine learning algorithms into production software. Experienced in designing, developing, testing, and maintaining robust software solutions within multidisciplinary R&D teams.

PROFESSIONAL EXPERIENCE

R&D Software Engineer

2022 – Today

Brest – France

INTRADYS

- Developed and maintained C++ software for real-time image processing in an industrial environment.
- Designed and developed high-performance software pipelines for filtering, registration, and 3D reconstruction.
- Developed, trained, and validated Deep Learning (nnUNet) models for automatic surgical instrument segmentation.
- Optimized algorithm performance under strict real-time and reliability constraints.
- Collaborated with multidisciplinary teams to translate business requirements into robust software solutions.

PhD - Towards a markerless CAOS system

2018 – 2024

Brest – France

IMT Atlantique / LATIM INSERM U1101

- Designed and developed a complete C++ software for real-time 6D pose estimation of surgical instruments (Point Pair Feature).
- Developed high-performance C++ algorithms for RGB-D point cloud processing, registration, and analysis.
- Designed, optimized, and validated algorithms under strict performance and robustness constraints.
- Designed experimental protocols and benchmarked algorithms on synthetic and clinical datasets.
- Co-invented a patented method for surgical instrument positioning verification using RGB-D cameras.
- Supervised laboratory sessions for simulator development on Unity.

Software Systems Engineering Intern

2018

Brest – France

IMT Atlantique

- Developed a C++ software application for real-time multimodal data fusion.
- Developed algorithms for 3D surface reconstruction and mesh processing.

EDUCATION

PhD - Towards a markerless Computer Assisted Orthopaedic Surgery system

2024

Brest – France

IMT Atlantique / LATIM INSERM U1101

Master's Degree in Computer Vision

2018

Alger – Algérie

University of Science and Technology Houari Boumediene (USTHB)

Bachelor's Degree in Software and Information Systems Engineering

2016

Alger – Algérie

University of Science and Technology Houari Boumediene (USTHB)